Siemens Energy
Global GmbH & Co KG

DAkkS Deutsche Akkreditierungsstelle D-PL-21472-01-00

Test report

Protection Class:Restricted

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Editor	Countersign	Date	Page
Bergner	Höhlein	04.07.2023	1 of 2

Address / customer
Savita oil Technologies Ltd.
66/67 Nariman Bhavan, Nariman point Mumbai 400021 India

sample reception

Distribution list
Savita

India

PO: 4200011909

SE GT SV PFS FN ML

Testing of the synthetic ester dielectric liquid Transol Synth. 100 from Supplier Savita, India

The synthetic ester Transol Synth.100 from Savita, India fulfils the requirements in the delivery state and for the oxidation stability after ageing according to IEC 61099:2010.

1 Cause of Study

The synthetic ester Transol Synth.100 from Savita, India has been delivered for testing and evaluation acc. to IEC 61099:2010.

2 Tests

The synthetic ester Transol Synth.100 from Savita, India has been tested in the delivered state and for ageing stability according to IEC 61099:2010

3 Results

The tested properties of synthetic EsterTransol Synth.100 from Savita, India in comparison with the limit values of IEC 61099:2010 are represented in table 1.

Table 1. Properties of the the synthetic ester Transol Synth.100 from Savita, Indiai the delivery state and after aging

Property	Test Method	Unit	2393-2023 Transol Synth.100	Limit value (IEC61099:2010)
Colour	ISO 2049:1996		0	<1
Purity	IEC 60422:2013		clear	clear
Neutralisation value[1]	IEC 62021-3:2014	mg KOH/g oil	<0,01	≤ 0,03
Diel. loss factor at 90°C	IEC 60247:2004		0,006	≤ 0,03
DC at 90°C	IEC 60247:2004	GΩm	15	Min.2
Water content	IEC 60814:1997	mg H_2O/ kg oil	3	≤ 200
Density at 20 °C	ISO 12185:1996	kg/m³	963,5	≤ 1000
Breakdown voltage	IEC 60156:2018	kV	73	>45
Pour point[1]	ISO 3016:2019	°C	-63	max. -40°C
Corrosive sulfur	IEC 62535:2008		non corrosive	negative
Total additives DBPC Irganox 1010	IEC 60666:2010	%	0,75 n.d. 0,75	-
Viscosity at 40°C	ASTM D 7042:2020	mm²/s	30,0	max 35mm²/s
Viscosity at -20°C	ASTM D 7042:2020	mm²/s	1446	max 3000mm²/s
Viscosity at 100°C	ASTM D 7042:2020	mm²/s	5,38	max.15
Oxidation stability 164h[1] Total acidity Sludge	IEC 61125:2018	mg/g oil %	0,10 <0,01	max. 0,3 max.0,01
PCB content	IEC 61619:1997	mg/kg oil	n.d.	n.d.
DBDS content	IEC 62697-1:2012	mg/kg oil	n.d.	n.d.
Flash Point[1]	ISO 2592:2018	°C	290	min.250°C
Fire point[1]	ISO 2592:2018	°C	316	min.300°C

[1] Not in the scope of accreditation
n.d.=not detectable

4 Evaluation

The synthetic ester Transol Synth.100 from Savita, India fulfils the requirements in the delivery state and for the oxidation stability after ageing according to IEC 61099: 2010.

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Testing insulating fluids

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